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SWAP Circuit Identity

Video: SWAP Circuit Identity
4 min

Graded Assignment: Assessment 12
Submitted

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Next item

# Assessment 12

Review Learning Objectives

coach

Ready to review what you've learned before starting the assignment? I'm here to help.

Help me practice Let's chat

**Assignment details**

**Due**
Dec 25, 11:59 PM IST

**Attempts**
Unlimited

Retry

**Submitted**
Dec 11, 3:00 AM IST

**Your grade**
To pass you need at least 80%. We keep your highest score.

100%

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**1.** What is the result of applying a Hadamard gate to a quantum state $|0\rangle$?

1 / 1 point

- $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$
- $|1\rangle$
- $|0\rangle$
- $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ ?

**Correct**
Correct! This is the expected result when applying a Hadamard gate to $|0\rangle$.

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**2.** Which of the following statements are true about the controlled insert gate?

1 / 1 point

- [x] It has a 4x4 matrix representation.

  **Correct**
  Correct! The controlled insert gate has a 4x4 matrix representation.

- [x] It can create entanglement between qubits.

  **Correct**
  Correct! The controlled insert gate can create entanglement.

- [ ] It is a single-qubit gate.
- [ ] It is equivalent to applying two Hadamard gates in sequence.

**3.** Which truth table corresponds to the controlled CZ gate?

1 / 1 point

- 00 -> 10, 01 -> 11, 10 -> 00, 11 -> 01
- 00 -> 01, 01 -> 00, 10 -> 11, 11 -> 10
- 00 -> 00, 01 -> 10, 10 -> 01, 11 -> 11
- 00 -> 00, 01 -> 01, 10 -> 10, 11 -> 00

**Correct**
Correct! This is the truth table for the controlled CZ gate.

**4.** What is the primary function of the control Y gate in quantum computing?

1 / 1 point

- To measure the state of a qubit without collapsing it.
- To flip the qubit state conditionally based on the control qubit.
- To create entanglement between two qubits.
- To reset a qubit to its ground state.

**Correct**
Correct! The control Y gate flips the qubit state conditionally based on the control qubit's state.

**5.** What is the main effect of applying a swap gate to a pair of qubits?

1 / 1 point

- It applies a Hadamard operation on the qubits.
- It measures the qubits and resets them.
- The quantum states of the two qubits are exchanged.
- It entangles the two qubits.

**Correct**
Correct! The primary function of a swap gate is to exchange the states between two qubits.